

YawDD+ — Frame-level Annotations for Accurate Yawn Recognition on Edge Platforms

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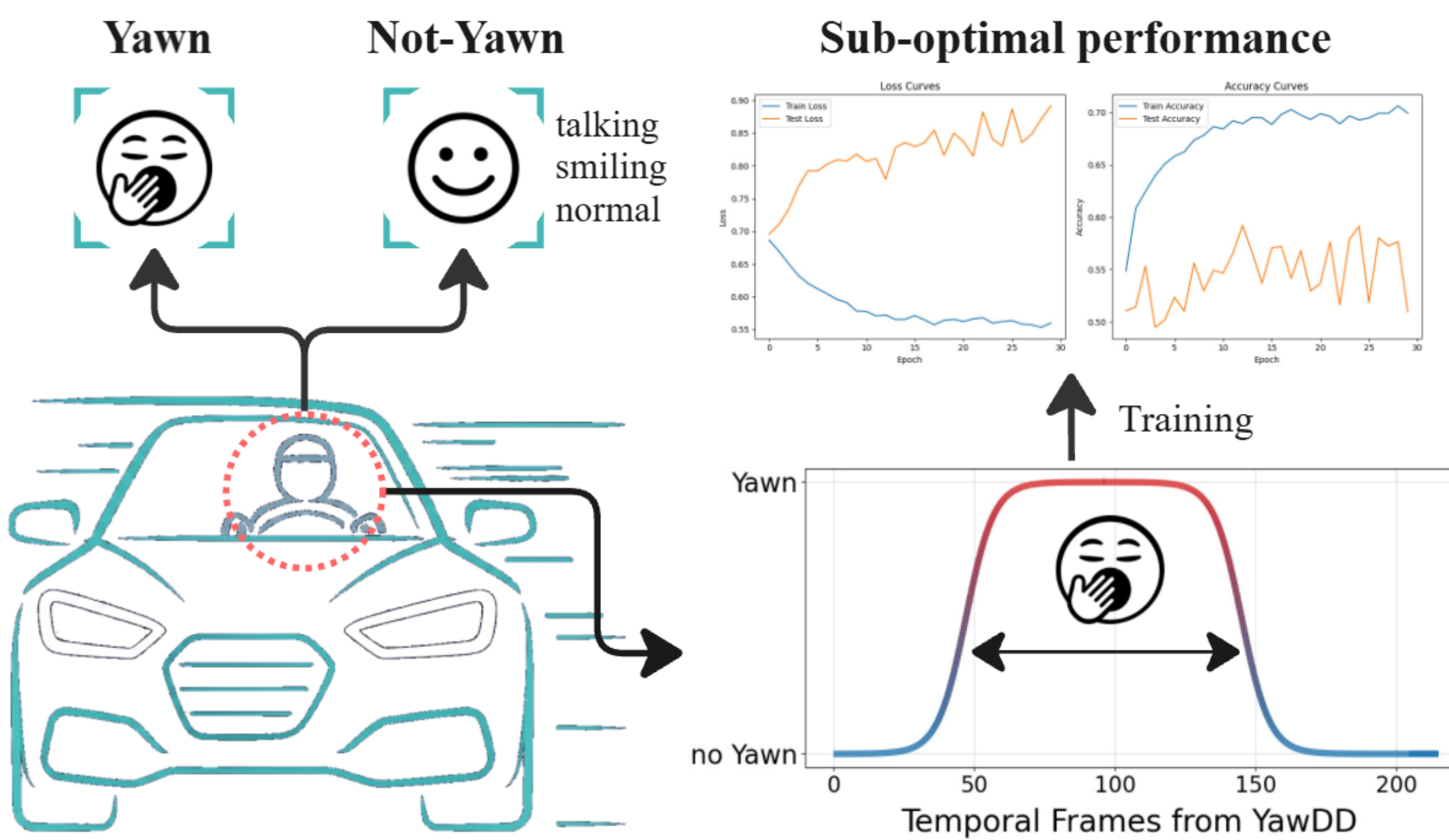


Motivation & Problem Statement

- Driver fatigue is a leading cause of accidents on roads.
- In 2017, US National Highway Traffic Safety Administration reported **91,000 crashes** involved drowsy drivers [1], estimating **50,000 injured and 800 deaths**.
- In 2019, e-survey of road users' attitudes reports that Austria has the highest driver fatigue percentage (**31.8%**) in Europe [2].

Problems in Existing Datasets

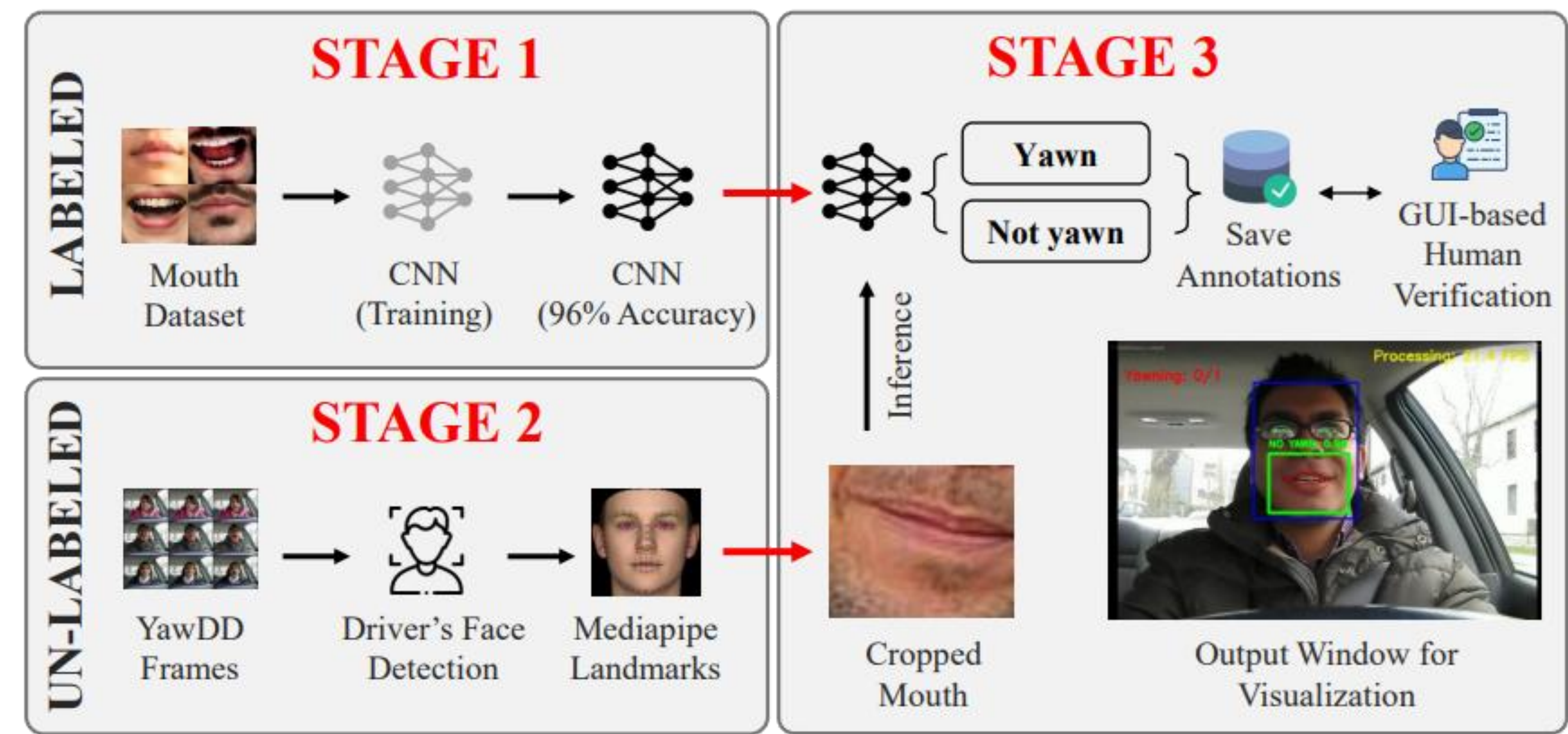
- Existing yawn datasets (YawDD [3]) contain systematic label noise.
- Noisy supervision:** Models learn from incorrectly labeled frames.
 - Expensive models:** Temporal ML architectures compensate for coarse video-level annotations; however, they introduce substantial latency on edge platforms which is highly risky for driver-monitoring applications.



The actual yawn episode occupies only a short temporal segment with transition phases.

Key Contribution — YawDD+ Annotations

- ★ **YawDD+** — High-quality frame-level annotations for fine-grained supervision for accurate driver yawn recognition.
- ★ A novel three-staged **semi-automated data annotation pipeline** using deep learning models with human-in-the-loop verification.
 - ★ A total of 287k images were annotated in less than a week.
≈22k [Yawn] — ≈264k [Not-yawn]



- ★ State-of-the-art performance comparison with existing studies on driver yawn recognition task on edge platforms.

Models	Accuracy / mAP	Edge Platform	Inference
Temporal ML Architectures			
MC3	84.62%	Jetson NANO	4.45 s
R3D	80.77%	Jetson NANO	4.62 s
SWIN	92.3%	Jetson NANO	10.79 s
Classification Models			
Two-stage CNN	93.83%	Raspberry Pi 4	96.3 ms
CNN	94.5%	Jetson NANO	166 ms
MobileNetv3	94%	Jetson Xavier TX2	45.4 ms
MNasNet (Ours)	99.34%	Jetson NANO	16.71 ms
Detection Models			
YOLOv5, YOLOv8	90.1%, 90.3%	—	—
YOLOv11 (Ours)	95.69%	Jetson NANO	35.7 ms

Outperforms all existing studies in both accuracy and inference speed on edge hardware.

Performance Analysis



Grad-CAM visualization of MNasNet focusing on driver's mouth region only.

YOLOv11 detection outputs for yawn and not-yawn subjects under various illuminations.



CONNECT

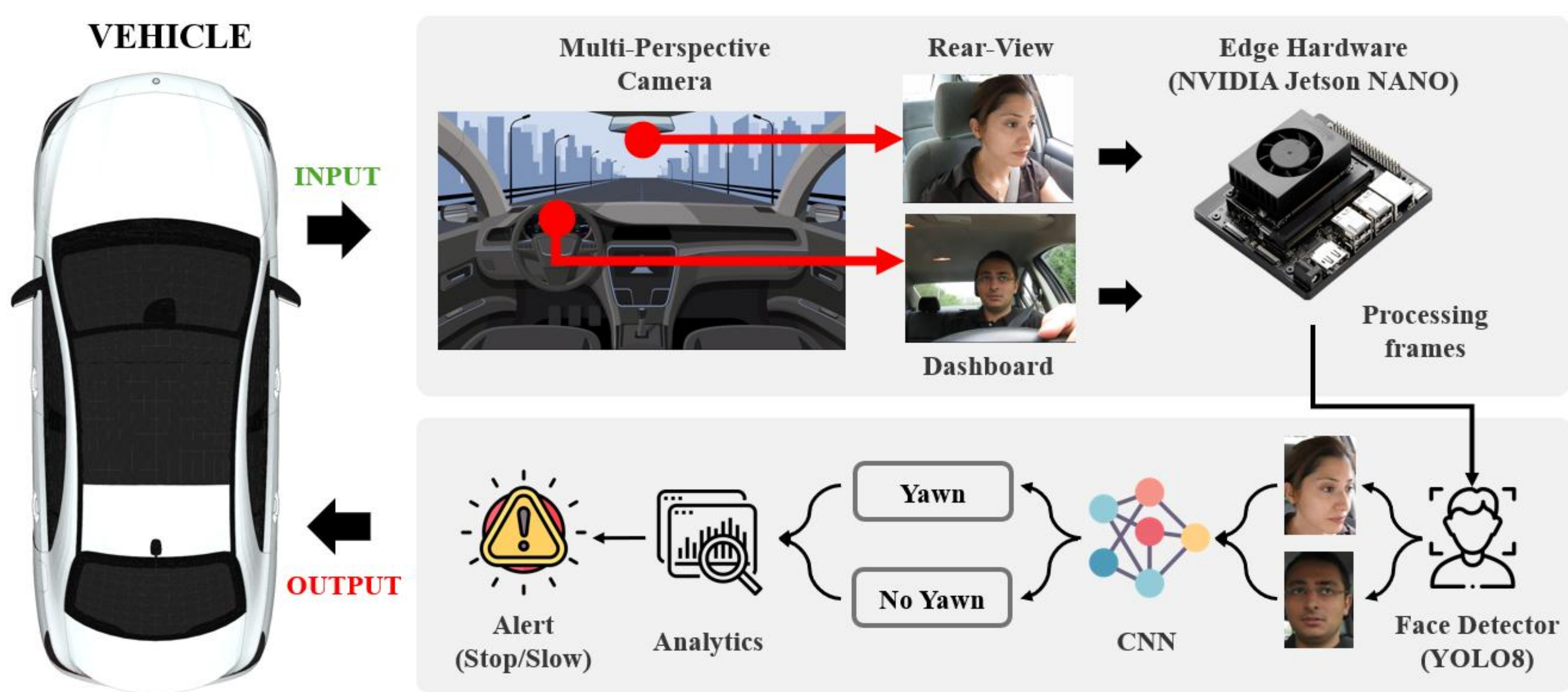


DATASET



PAPER

Driver Yawn Recognition Pipeline on Vehicles



On-Device Learning without quantization or pruning.

MNasNet

- ★ 13.93 min/epoch on NANO.
- ★ 8.69 min/epoch on AGX.

YOLOv11

- ★ 85.8 min/epoch on NANO.
- ★ 41.4 min/epoch on AGX.

None of the existing studies demonstrated on-device training for driver yawn recognition.

Conclusion: YawDD+ provides precise frame-level supervision that enables lightweight models to outperform expensive temporal architectures while running entirely on edge hardware without quantization or pruning.

References

[1] <https://www.nhtsa.gov/risky-driving/drowsy-driving>
[2] https://road-safety.transport.ec.europa.eu/system/files/2021-07/road_safety_thematic_report_fatigue_tc_final.pdf
[3] Abtahi, S., Omidyeganeh, M., Shirmohammadi, S. and Hariri, B., 2014, March. YawDD: A yawning detection dataset. In Proceedings of the 5th ACM multimedia systems conference (pp. 24-28).

Acknowledgements. This work was funding by EU-MSCA COFUND project CRYSTALLINE, Austrian Research Promotion Agency (AIM AT Stiftungsprofessur für Edge AI), and partially funded by the Austrian Science Fund (FWF) 10.55776/COE12.

